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EDUCATION

University of Alberta

Bachelor of Science in Computer Science

Edmonton, AB

2025 – Present

PROJECTS

PocketPal – Expense Tracker Web App | *Python, Flask, SQLite, Jinja2, Chart.js*

Live Demo

- Built a full-stack personal finance application with dual authentication: secure email/password login and Google OAuth 2.0 sign-in integration.
- Implemented robust password validation enforcing complexity requirements, with encrypted credential storage and protection against common attack vectors.
- Developed full CRUD expense management with multi-field filtering including date range, category, amount via RESTful Flask routes, enabling efficient retrieval across large transaction histories.
- Engineered an analytics dashboard with interactive bar and line charts (Chart.js) visualizing monthly spending trends and category level breakdowns, giving users actionable financial insights.
- Designed a normalized SQLite database schema with foreign key constraints and server-side input validation to ensure data integrity and prevent malformed entries across all operations.
- Built an account management module allowing users to upload and update profile pictures, change passwords with re-authentication confirmation, and permanently delete their account and associated data.
- Deployed to Render for public access, configuring environment variables and production-safe settings for secure cloud hosting.

Ledger – Job Application Tracker | *Python, Flask, PostgreSQL, Jinja2*

Live Demo

- Built a full-stack job application tracker featuring a drag-and-drop Kanban board across four pipeline stages (Applied, Interview, Offer, Rejected), with real time status persistence via RESTful Flask routes.
- Migrated the data layer from SQLite to a Supabase-hosted PostgreSQL database, implementing incremental schema migrations to preserve existing user data across iterative feature releases.
- Engineered secure authentication combining Supabase Auth (email/password with server-side session management) and Google OAuth 2.0, supporting login via either username or email through custom identity resolution logic.
- Developed application-lifecycle features including interview date and application deadline tracking with dynamic urgency indicators, soft-delete archiving with restore capability, and one-click CSV export for full data portability.
- Built an account management module supporting profile picture uploads, email and password updates, and permanent account deletion integrated with Supabase's administrative API.
- Implemented a fully responsive, mobile first interface with a swipeable touch-friendly board layout and persistent dark mode, deployed to Render for public access.

AI Spending Assistant | *React*

Live Demo

- Built a React chat interface that lets users ask natural-language questions about their PocketPal expense data, integrated with an LLM API to generate conversational responses.
- Designed the backend to precompute all financial calculations (totals, category breakdowns, monthly averages, min/max transactions) in Python before passing them to the LLM, eliminating AI arithmetic errors and ensuring 100% accurate financial figures.
- Implemented token based authentication using signed, expiring tokens instead of session cookies, resolving cross-origin authentication failures caused by browser third-party cookie restrictions and ensuring reliable login across devices and browsers.
- Built a RESTful Flask API endpoint that dynamically queries and categorizes a user's live expense data on each request, supporting follow-up questions with contextual accuracy.
- Deployed frontend and backend as independent services on Render, configuring CORS and cross origin request handling for secure client-server communication.

TECHNICAL SKILLS

Languages: Python, C, JavaScript, HTML/CSS, SQL

Frameworks & Tool: React, Flask, Linux

Developer Tools: GitHub, Git, VS Code

Libraries: Pandas, NumPy, Matplotlib, Chart.js, SQLite

Relevant coursework: OOP, Data Structures & Algorithms, Machine Learning, Operating Systems